

WONJOON HWANG

Seattle, WA | wh4570@uw.edu | [LinkedIn](#) | [GitHub](#) | [Portfolio](#)

SUMMARY

M.S. in Data Science candidate and Databricks Student Fellow with experience building ML and data workflows using Python, SQL, Spark, and Databricks. Research and project experience spans federated ML, predictive modeling, anomaly detection, model evaluation, and end-to-end data pipelines across environmental and geospatial datasets.

EDUCATION

UNIVERSITY OF WASHINGTON

M.S., Data Science (GPA: 3.86)

Sep 2025 – Mar 2027

Seattle, WA

UNIVERSITY OF GEORGIA

B.S., Data Science | Minors: Statistics & Computer Science (Major GPA: 3.68)

Aug 2022 – May 2025

Athens, GA

- **Achievements:** Dean's List: Spring 2023, Fall 2023, Spring 2024

EXPERIENCE

Databricks | Databricks Student Fellow

Jun 2026 – Present

- Build end-to-end **Python, SQL, Spark, and Databricks** workflows for ingestion, transformation, validation, analytics, and ML.
- Develop reusable technical demos of lakehouse and data-engineering workflows for mixed technical audiences.
- Selected as a **Databricks Student Fellow** after a **1st-place UW hackathon** finish; presented TerraCast at the Data + AI Summit.

Translation Commons | Volunteer Data Engineer

Jun 2026 – Sep 2026

- Shipped **21 production fixes** across **9 language groups**, correcting parent-child relationships and improving shared dataset consistency.
- Built a scope-validation tool that detected **95 hierarchy inconsistencies**, catching structural data issues before downstream use.
- Added **9 unit tests** for parent-override behavior and issue detection to validate future hierarchy updates.

University of Georgia | Research Assistant - Federated & Distributed ML

Sep 2023 – Jan 2024

- Implemented and evaluated **FedSVD** and **FedPower** in **Python** and **TensorFlow**, simulating decentralized ML across multiple clients with **non-IID data** to study performance under heterogeneous distributions.
- Benchmarked federated methods against **centralized** and **baseline approaches**; analyzed **convergence, accuracy, and communication/computation trade-offs** to compare distributed training strategies.

PROJECTS

TerraCast

Mar 2026

Databricks x UW Paul G. Allen School Hackathon - 1st Place

- Built an end-to-end **Python, SQL, Spark, and Databricks** pipeline integrating county-level crop yield, weather, and climate data; trained a **LightGBM** model achieving **0.88 cross-validated R²**.
- Applied **residual-based anomaly detection** to identify **31 abnormal county-year outcomes (0.45%)**, focusing investigation on potential environmental and operational drivers.
- Presented yield and risk analytics to **Corteva** at the Data + AI Summit; representatives noted close agreement between predicted and actual yields.

Water Quality Prediction

Jan 2026 – Mar 2026

EY Open Science AI & Data Challenge, Snowflake Collaboration

- Integrated and validated **9,319 water-quality observations** with **205 satellite, climate, hydrology, and geospatial features** from Landsat, TerraClimate, ERA5, and HydroRIVERS.
- Engineered environmental features and designed **spatial cross-validation** to reduce leakage and evaluate **generalization across unseen regions**.
- Ranked **66th of 2,500+ teams** on the final leaderboard in the 2026 EY Open Science AI & Data Challenge.

Darin

Jun 2026 – Present

Product Developer | Pre-Launch Product

- Developed and maintained TypeScript application workflows for a private family memory application.
- Modularized a **1,713-line** core editor to **205 lines** and removed **1,071** hard-coded UI strings while maintaining parity across **5 languages**.

WatchDawg

Sep 2025 – Dec 2025

University of Washington

- Developed **Python** and **SQL** workflows with **Dash** and **DuckDB** to transform, aggregate, and visualize public-safety data.
- Built interactive **geospatial and time-series views** to explore incident patterns by location, time, precinct, and incident type.
- Added **LightGBM Poisson forecasting** and drill-down analysis to surface changes in local incident patterns.

SKILLS

- **Programming:** Python, SQL, R, Java, TypeScript
- **ML/Statistics:** TensorFlow, PyTorch, scikit-learn, LightGBM, XGBoost, federated ML, model evaluation, statistical modeling
- **Data & Analytics:** Spark, PySpark, Databricks, Pandas, NumPy, ETL/data pipelines, data validation, geospatial analysis, anomaly detection, forecasting
- **Databases:** PostgreSQL, MySQL, DuckDB, Snowflake, Supabase
- **Cloud & Systems:** AWS, Google Cloud, Git, Docker, REST APIs, CI/CD, backend systems, automation workflows